DSA 8420 Spring 2025
Homework 11

Due April 22, 2025

Set up a (mixed) integer linear programming model for each of the following problems. Make sure you state the definition of the variables. Do not solve the problem.

1. (10 points) A chair manufacturer makes three different types of chairs, each of which must go through sanding, staining, and varnishing. In addition, the model with the vinyl-covered back and seat must go through an upholstering process. Table 1 gives the time required for each operation on each type of chair, the available time for each operation in hours per month, and the profit per chair for each model. How many chairs of each type should be made to maximize the total profit?

Table 1

Model	Sanding (hours)	Staining (hours)	Varnishing (hours)	Upholstering (hours)	Profit ($)
A – solid back and seat	1.0	0.5	0.7	0	10
B – ladder back, solid seat	1.2	0.5	0.7	0	13
C – vinyl-covered back and seat	0.7	0.3	0.3	0.7	8
Total time available per month	600	300	300	140	

2. (10 points) A company is considering several investment opportunities, each of which differs in the initial capital required and cannot be invested repeatedly. The accounting department has done a thorough analysis of each of these investments and has estimated the long-term profit of each. The initial capital required and estimated profits (in millions of dollars) are summarized in Table 2. Investments 1 and 4 are considered high-risk investments and management has decided to invest in at most one of these. In addition, Investment 6 is contingent upon also investing in Investment 3, that is, Investment 6 is invested only if Investment 3 is invested. If $120 million of initial capital is available, formulate an integer programming model to determine the optimal investment strategy.

Table 2

Investment	Initial capital	Estimated profit
1	26	18
2	34	12
3	18	7
4	45	24
5	31	11
6	39	15
7	23	9
8	13	6

3. (10 points) A city is trying to establish a municipal emergency ambulance service that can adequately service all sections of the city. The planning committee has established eight potential sites for the emergency service facilities. However, due to time and distance requirements, each potential site can provide coverage to only a subset of the city's six sections. Table 3 summarizes the sections for which each site provides coverage.

   Formulate an integer programming model for determining the fewest number of facilities that will provide all sections of the city with adequate coverage.

4. (10 points) Distributed oil deposits are often tapped by drilling from a central site. BlackGold Corp. is considering two potential drilling

Table 3

Site	Sections of city covered
1	A, B, E
2	C, D
3	B, C, D
4	A, D, F
5	B, F
6	A, D, E, F
7	A, C, E
8	B, D, F

sites for reaching four targets (possible deposits). Table 4 provides the preparation costs (in millions of dollars) at each of the two sites and the cost of drilling from each site to each deposit. Formulate the problem of choosing which sites to open and which sites should tap each deposit as an integer program.

Table 4

	Drilling cost to target				Preparation
Site	1	2	3	4	cost
1	2	1	8	5	5
2	4	6	3	1	6

5. (10 points) Each day at the Graphic Arts Co. the press operator is given a list of jobs to be done during the day. He must determine the order in which he does the jobs based on the amount of time it takes to change from one job setup to the next. He will arrange the jobs in an order which minimizes the total setup time. Assume that each day he starts the press from a rest state and returns it to that state at the end of the day. Suppose on a particular day he must do six jobs for which he estimates the changeover times given in Table 5. What schedule of jobs should the operator use? (Feel free to use c_{ij} to represent the coefficients in the table if necessary.)

Table 5

	To job j (minutes)						
From job i	1	2	3	4	5	6	rest
1	0	10	5	15	10	20	5
2	10	0	10	10	20	15	10
3	5	5	0	5	10	10	15
4	8	10	3	0	9	14	10
5	4	7	8	6	0	10	10
6	10	5	10	15	10	0	8
rest	7	7	9	12	10	8	0